

Water Transport

Via pipelines

Causes, Effects, and Solutions



Master thesis



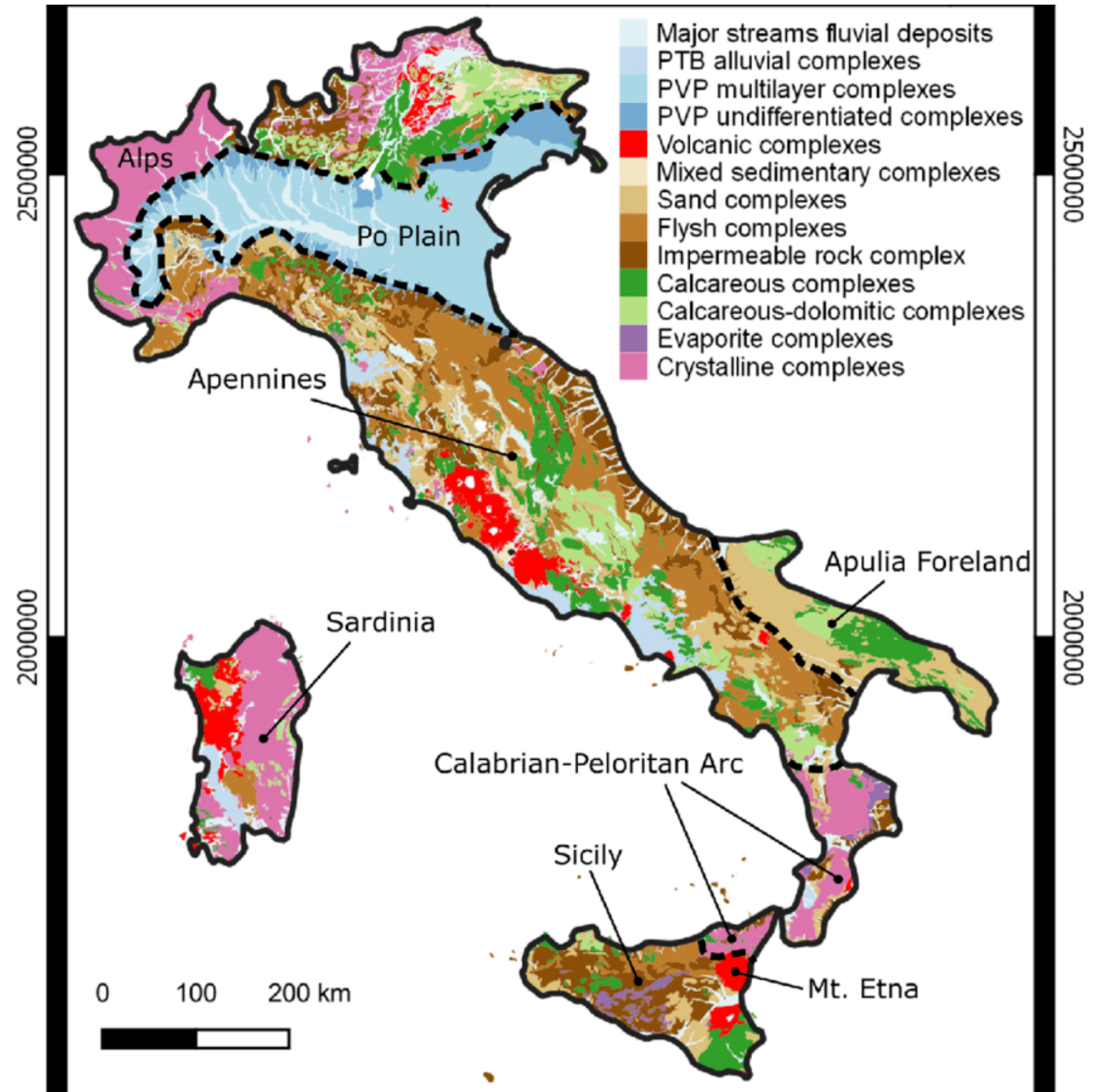
Transforming multimodal open data into decision making



*What is water
transportation
through pipelines?*

Introduction

Italy's water system relies on **Alpine** and **Apennine basins**, dominated by the **Po River** in the north and karst aquifers in the south. However, the country faces a **critical transition** as **climate change** intensifies droughts and **aging infrastructure** results in water losses **exceeding 40%**. To combat this, the efforts aim to transform obsolete pipelines into **smart, resilient networks** that can safely transport "blue gold" across the peninsula.





 Evidence of
the problem

Italian reality

Problem setting

Some of Italy biggest concerns in the field

Transport inefficiencies

In Italy, the inefficiency of water distribution infrastructure is a major systemic issue closely linked to long-distance transport and resource sustainability. (ISTAT)

Approximately **3.4 billion cubic meters annually** — is equivalent to the annual water consumption of **about 43.4 million people**. (ANSA.it)

Infrastructure and Environmental factors

The aging infrastructure contributes to this problem: a large share of Italian water networks were installed decades ago, increasing leak risk and maintenance costs. (length **(425,000–450,000 km)**) (Blue Book, 2025)

Climate-related hazards compound the problem. in 2024, approximately **94.5 % of municipalities were at risk of landslides, floods, coastal erosion, or avalanches, and landslide-prone areas have expanded to cover 23 % of the national territory** (Isprambiente)



Gap

Literature review



*What has
been done?*

Relevant ideas

- Urban water distribution system for smart cities, through IoT sensors and external data (Figueiredo et al., 2021)
- Floods prediction with social media inputs such as Twitter (Donratanapat et al., 2020)
- HydroSense 2.0, real-time water quality surveillance (Mattihalli et al., 2025)
- Reviews on blockchain-based systems for ground water and wastewater systems (Mithal & Thankachan, 2025)

Relevant ideas (from other fields)

- Multi source data fusion framework based on ontology structures for structured, semi-structured and unstructured data types (Liu et al., 2019)
- Introducing flexible non-linear ensembles using directed acyclic graphs (DAG's) for predictive maintenance (Samadder et al., 2025)
- GNNs focused on stock forecast using multi source heterogeneous data (Li et al., 2022)

Preliminary exploration

Data from **ISTAT** and **Eurostat** provide a comprehensive socio-economic overview via long-term time series. The granularity of these datasets is generally restricted to annual or biennial reporting cycles, facilitating high-level national context analysis.



Dati.gov.it and ARPA entities

Dati gov is a site that allows many Italian institutions to host their data in its website for easier access, the site serves as an integrator.

Agenzia Regionale Protezione Ambiente (ARPA) is an agency divided in the Italian administrative regions serving the environment protection.



Open data regional sites

Each Italian region has an open data website which contains useful geodata, water basins, hydrology, pipeline architecture, water distribution schemas and more.



Approach

Potential sources

Environmental risk sites and Patent data

Programs like **ERA5** from **Copernicus**, **IdroGEO** from **Isprambiente**, **CFTI landslides** from the national institute of geophysics and vulcanology (**INGV**) and **Copernicus crops data**.

Patent data could help obtain technical knowledge about the physical factors that affect the pipeline systems at hand.



Dati.gov.it

Meta data of the datasets was collected for water related and sewage related data with the website's API.

Regional open data websites

Data was collected from the following regional sites: Calabria, Campania, Lazio, Puglia, Sicilia, Umbria.

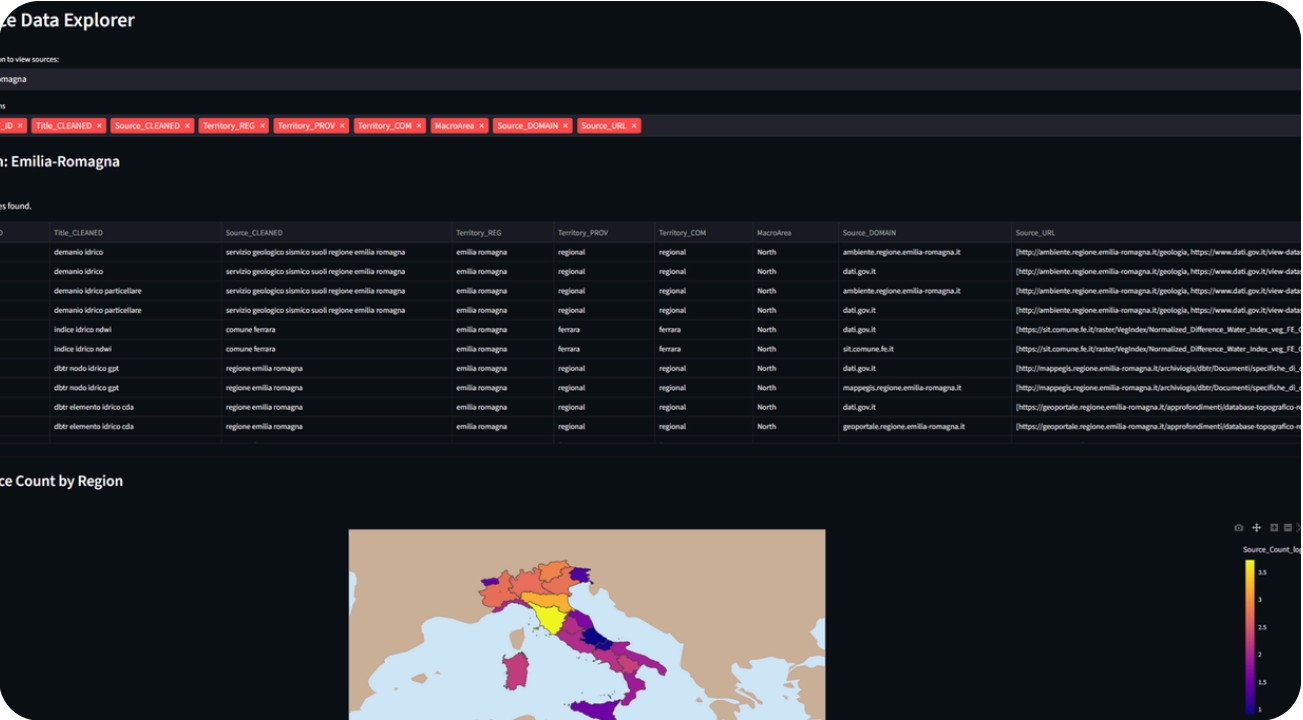
The reasoning was to complement the dati.gov.it integrator for regions that didn't have many datasets extracted.

Sites like Basilicata were inaccessible for architecture changes currently in their service

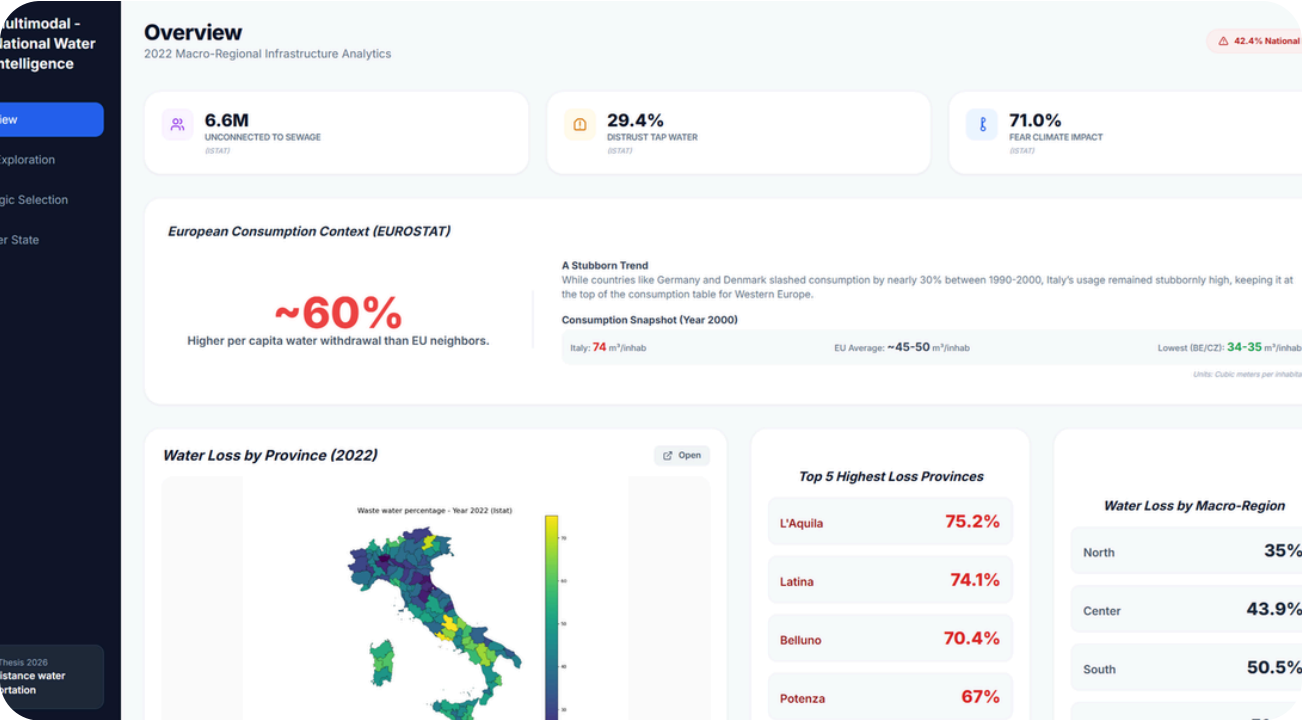


Approach

Collecting the data



Building the
dashboards



Giving meaning to the metadata collection

Two dashboards were created with different objectives in mind.

- The first (LEFT) delving only into the data and the insights found.
- The second (RIGHT) delving into a more presentation oriented board, meant to display the problem and what has been found.

Analysis
construction

Research question & objectives

Main Research question

How can multimodal open data and patent data be used to reduce inefficiencies in water transportation systems through pipelines?

Secondary objectives

To identify and systematically catalog diverse repositories of multimodal open data and global patent databases relevant to the Italian water transport system.

Secondary objectives

To establish a comprehensive methodology for the integration and harmonization of heterogeneous datasets, ensuring interoperability between various knowledge fields and from structured to unstructured data.

Secondary objectives

To design a data-driven framework that utilizes integrated insights to support the prioritization of infrastructure interventions and the optimization of resource allocation.





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